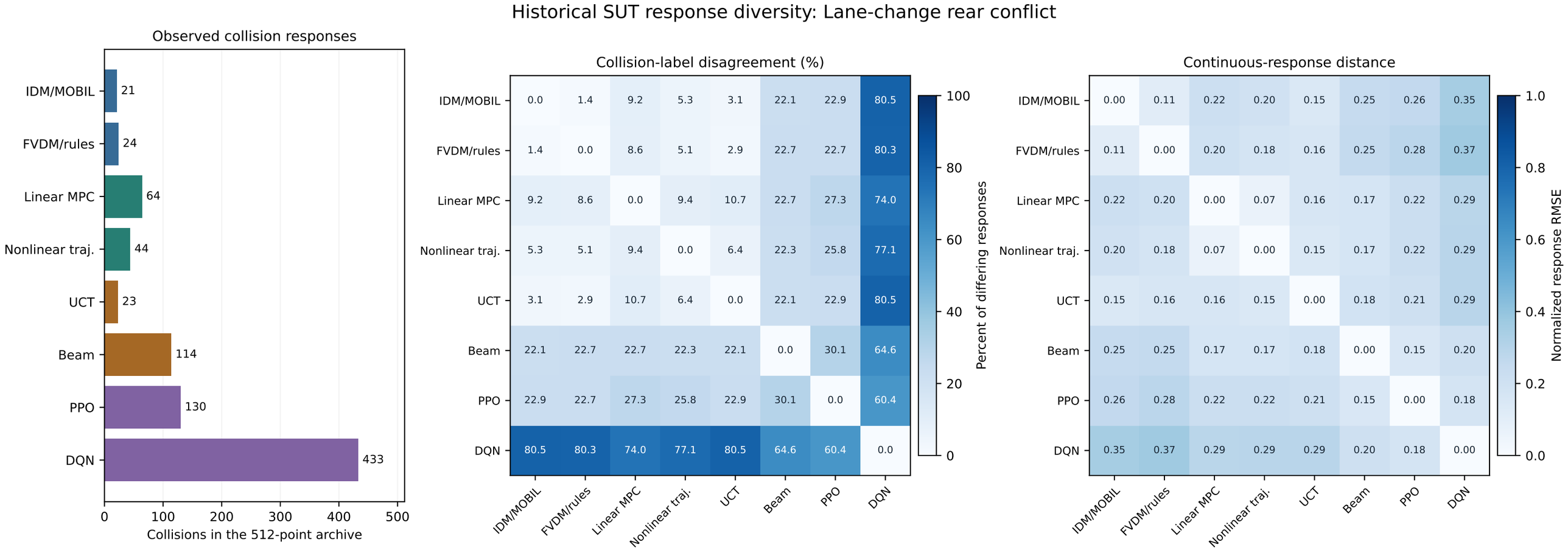
Historical SUT response diversity: Lane-change rear conflict
Observed collision responses
IDM/MOBIL
FVDM/rules
Linear MPC
Nonlinear traj.
UCT
Beam
PPO
DQN
21
24
64
44
23
114
130
433
0
100
200
300
400
500
Collisions in the 512-point archive
Collision-label disagreement (%)
0.0 1.4 9.2 5.3 3.1 22.1 22.9 80.5
1.4 0.0 8.6 5.1 2.9 22.7 22.7 80.3
9.2 8.6 0.0 9.4 10.7 22.7 27.3 74.0
5.3 5.1 9.4 0.0 6.4 22.3 25.8 77.1
3.1 2.9 10.7 6.4 0.0 22.1 22.9 80.5
22.1 22.7 22.7 22.3 22.1 0.0 30.1 64.6
22.9 22.7 27.3 25.8 22.9 30.1 0.0 60.4
80.5 80.3 74.0 77.1 80.5 64.6 60.4 0.0
Percent of differing responses
0
20
40
60
80
100
Continuous-response distance
0.00 0.11 0.22 0.20 0.15 0.25 0.26 0.35
0.11 0.00 0.20 0.18 0.16 0.25 0.28 0.37
0.22 0.20 0.00 0.07 0.16 0.17 0.22 0.29
0.20 0.18 0.07 0.00 0.15 0.17 0.22 0.29
0.15 0.16 0.16 0.15 0.00 0.18 0.21 0.29
0.25 0.25 0.17 0.17 0.18 0.00 0.15 0.20
0.26 0.28 0.22 0.22 0.21 0.15 0.00 0.18
0.35 0.37 0.29 0.29 0.29 0.20 0.18 0.00
Normalized response RMSE
0.0
0.2
0.4
0.6
0.8
1.0